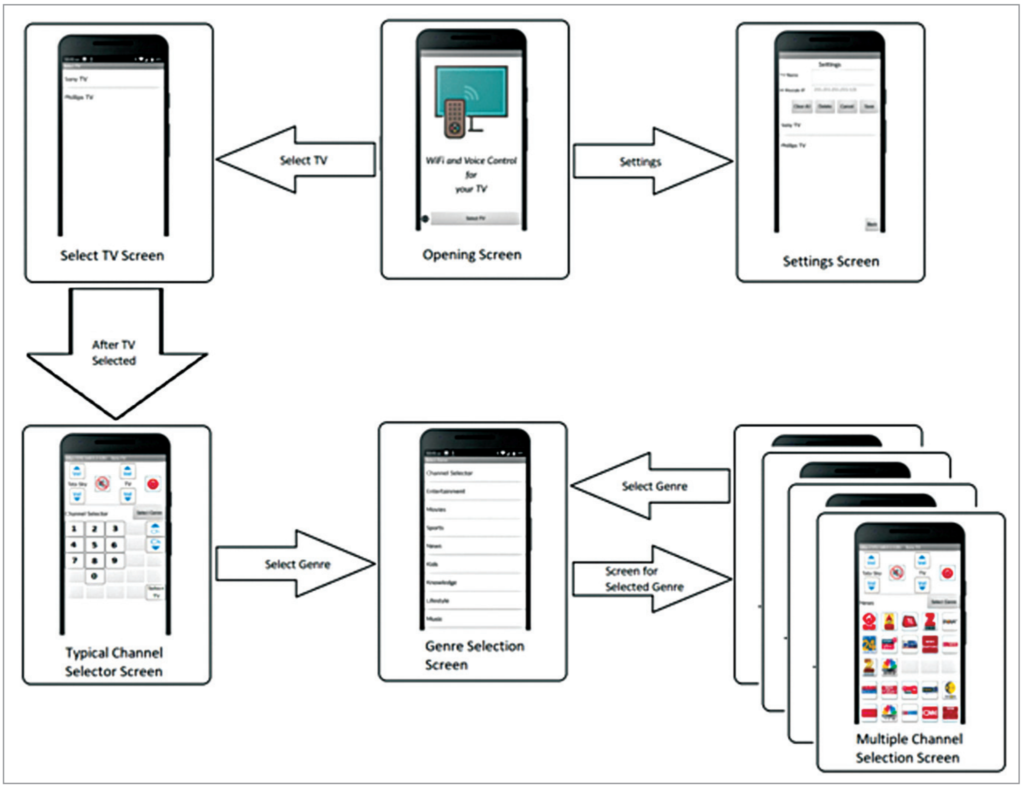

Fig. 5: Screenshots of sequence of operations of the app

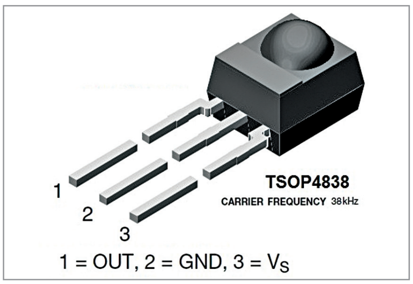

Fig. 6: TSOP4838 IR receiver

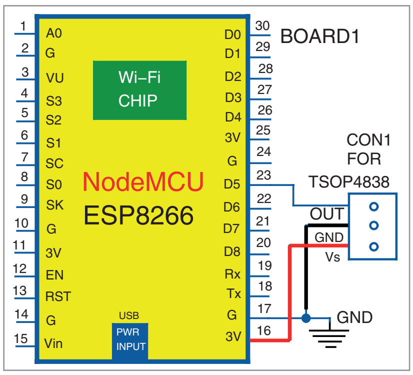

Fig. 7: Circuit diagram of IR decoder

customising and app development are given in the source code folder.)

The app is designed for a TataSky set-top box. Users using other brand set-top boxes will have to modify (change the channel numbers) the AIA file and rebuild the app's apk file. New users of MIT App Inventor 2 should read the article on MIT App Inventor available at *https://www.electronicsforu.com/electronics-projects/designing-android-app-with-bluetooth-module*

Decoding the TV/set-top box remotes

Before building IR-Blaster for the Wi-Fi remote, we need to first decode the signals transmitted for each key press. These keys are unique for a brand of TV remote and set-top box remote.

Since decoding the remotes is a one-time job, we may build this circuit on a (temporary) breadboard. We need a NodeMCU and an IR receiver IC. Many IR receiver ICs like the TSOP4838, TSOP1738, and TSOP1838 are available, and anyone will serve our purpose. Commonly available TSOP4838 has been used in the prototype. It is a 3-pin device having an IR photodiode and a 38MHz demodulator (Fig. 6).

The complete circuit of the IR decoder is shown in Fig. 7. The Out pin is connected to pin D5 (GPIO 14) of NodeMCU (Board1), Vs pin is connected to 3V or 3.3V pin, and Gnd to Gnd pin of the NodeMCU.

Connect the NodeMCU to your PC/laptop and start the Arduino IDE. Select NodeMCU as your board. Load the *IRremoteESP8266* library using Library Manager option. This library helps in decoding and encoding the IR signals.

Go to the Examples option and select *IRrecvDumpV2*. This will open the *IRrecvDumpV2.ino* file. Compile and load this sketch to your NodeMCU. Start the serial monitor from the Tools menu. Select 115200 baud as the speed for data transfer.

Point your remote towards the TSOP4838 IR receiver and press a button. The program will decode the IR signal and display the decoded signal for that particular key on the serial monitor. The respective code (HEX value) and encoding type (protocol) will be displayed. Please note down the name of the key, its code and encoding (Fig. 8). This way, note down codes for all the keys of your remote.

The lists of codes for TataSky set-top box and Sony TV used for the prototype are given in Fig. 9.

Building the IR-Blaster

The IR-Blaster part of Wi-Fi remote is built using another NodeMCU (Board2). The circuit diagram for the IR-Blaster is shown in Fig. 10. To give a stronger signal, the LEDs are driven by a 5V supply via two 2N2222 NPN transistors. The version shown has two IR LEDs (IRTX1 and IRTX2) to emit a very strong IR signal, though one IR LED is enough for a small room. The second IR LED should be added only if one IR LED is not enough.

The IR-Blaster should be housed in a suitable enclosure and mounted